

FINAL PROJECT

DevFlow CRM

CRM Built Specifically for Software Houses

Full-Stack Development Internship — U Devs 2026

Official Updated Project Brief / Assignment Specification

This updated brief preserves the original DevFlow CRM project scope and adds the official submission, verification, certification, approval, and originality instructions requested by U Devs.

1. Project Overview

DevFlow CRM is a production-style CRM designed specifically for software houses. The project connects leads and clients with projects, requirements, tasks, documents, invoices, support, teams, and internal operations through a role-based dashboard.

Technology: PERN + JWT + Redux Toolkit + Bootstrap/Tailwind + Material Icons. The implementation standard includes responsive UI, reusable components, secure REST APIs, relational database design, RBAC, validation, error handling, auditability, and Git-based development.

2. Required Technology & Core Scope

Area	Required Scope
Frontend	React.js + Vite, responsive UI, reusable components
Backend	Node.js + Express.js REST API, controllers, services and middleware
Database	PostgreSQL + Sequelize models, migrations and relationships
Authentication	JWT, access/refresh tokens, protected routes, password reset and email verification
State	Redux Toolkit for authentication, CRM entities and application state
CRM	Leads, prospects, clients, contacts, companies, deals and sales pipeline
Projects	Projects, milestones, tasks, subtasks, assignments and deadlines
Client Portal	Client projects, requirements, tickets, messages, documents and invoices
Requirement Vault	Versioning, attachments, comments, approvals and change history
Operations	Notifications, search, filters, pagination, activity log and audit trail
Testing	API testing, validation, RBAC testing and critical-flow verification

3. Email Verification Requirement — NodeMailer

The project must implement email verification using **NodeMailer** (or an equivalent SMTP-based Node.js email service) as part of the authentication workflow. Verification should include a verification token/link, email delivery configuration through environment variables, token validation, resend-verification handling, and appropriate success/error responses.

Recommended environment variables include SMTP_HOST, SMTP_USER, SMTP_PASS and CLIENT_URL. SMTP credentials and production secrets must never be committed to GitHub.

4. GitHub Submission & Verification

Every candidate must submit the completed project through a GitHub repository. The repository should contain the required frontend and backend source code, database setup/migrations, README, environment-variable documentation, API collection where applicable, and a clean commit history.

Submission should be made through GitHub using a repository that can be reviewed by U Devs. Candidates are expected to demonstrate their own implementation, development history, and working project during evaluation.

5. Mandatory Project Submission Rule

IMPORTANT — CERTIFICATE ELIGIBILITY: Only candidates who submit the completed project for evaluation are eligible to receive an internship/project completion certificate. A candidate who does not submit the required project will not receive a certificate.

Certificate issuance is subject to project submission and the organization's verification/evaluation process. Submission of a repository does not automatically guarantee certification if the required project scope or verification requirements are not met.

6. Original Work & No-Copyright/Plagiarism Instruction

All submitted work must be original and created by the candidate. Candidates must not copy another candidate's repository, source code, project files, credentials, documentation, screenshots, or other protected work and submit it as their own.

Candidates may use standard libraries, frameworks, official documentation, tutorials, and properly licensed resources where permitted, but they remain responsible for understanding and implementing their submitted work. Do not submit copyrighted third-party material without the appropriate rights or license. Any third-party assets or code used must comply with their applicable license and should be properly attributed where required.

7. Approval & Authorization

Approved by:

Usama Aslam	Founder & CEO, U Devs
Warda Fatima	Co-Founder, U Devs

Organization: U Devs SMC PVT Limited

Project: DevFlow CRM — Full-Stack Development Internship Final Project

Project Window: 25 August 2026 to 02 September 2026

8. Final Submission Checklist

- GitHub repository containing the complete project.
- React/Vite frontend and Node.js/Express backend.
- PostgreSQL database schema, Sequelize models/migrations and seed data.
- JWT authentication, RBAC and email verification using NodeMailer.
- Mandatory CRM, project management, Requirement Vault and client portal functionality.
- README with setup instructions, environment variables and run commands.
- API collection and testing evidence where applicable.
- Screenshots or demo evidence of major workflows.

- Clean Git history and no exposed passwords, SMTP credentials, API keys or production secrets.
- Original implementation with no plagiarism or unauthorized copyrighted material.

Official Instruction

No project submission = No certificate. Only candidates who submit the required final project through the designated GitHub submission process and complete the required verification/evaluation are eligible for certification.

Prepared / Issued By: Usama Aslam — Founder & CEO, U Devs

Approved By: Usama Aslam — Founder & CEO, U Devs | Warda Fatima — Co-Founder, U Devs

U Devs SMC PVT Limited | Software Solutions & Digital Services | 2026